

# POE / Semiconductor / EMT Project Catalogue

Student-ready project ideas — basic, intermediate, IoT, advanced electronics, semiconductor, communication and control systems

**Suggested listing note:** Projects can be adapted to Arduino, ESP32, STM32, Raspberry Pi, PIC, 8051 or other suitable platforms depending on requirements. Final implementation, components, documentation and testing scope can be customized.

## 1. Basic Sensor & Detection Projects

| No. | Project                           | Typical implementation / key modules |
|-----|-----------------------------------|--------------------------------------|
| 1   | Ultrasonic Radar / Object Scanner | HC-SR04, servo, Arduino              |
| 2   | LPG / Propane Gas Detector        | MQ-6, buzzer, Arduino                |
| 3   | Rain Detection System             | Rain sensor, relay, alert            |
| 4   | Soil Moisture Detection System    | Moisture sensor, pump control        |
| 5   | IR Obstacle Detection Sensor      | IR sensor, LED/buzzer                |
| 6   | Ultrasonic Distance Measurement   | HC-SR04, LCD/OLED                    |
| 7   | Weather Monitoring System         | Temperature, humidity, pressure      |
| 8   | Season Detection System           | Temperature/light/humidity sensing   |
| 9   | Air Quality Index Monitor         | MQ sensors, display                  |
| 10  | Smoke Detector & Alarm            | Smoke sensor, buzzer                 |
| 11  | Multi-Gas Detector                | MQ-2/MQ-5/MQ-6                       |
| 12  | Room Temperature Monitor          | LM35/DHT11/DHT22                     |
| 13  | RFID Security System              | RFID reader, servo/relay             |
| 14  | Home Intruder Alarm               | PIR, magnetic switch, buzzer         |
| 15  | Automatic Street Light            | LDR, transistor/MOSFET, LEDs         |
| 16  | Water Level Indicator             | Level probes/ultrasonic sensor       |
| 17  | Automatic Hand Sanitizer          | IR sensor, pump                      |
| 18  | Touchless Doorbell                | IR/proximity sensor, buzzer          |
| 19  | Digital Visitor Counter           | IR pair, 7-segment/LCD               |
| 20  | Flame Detection Alarm             | Flame sensor, buzzer/relay           |

## 2. Intermediate Embedded & Automation Projects

| No. | Project                             | Typical implementation / key modules |
|-----|-------------------------------------|--------------------------------------|
| 21  | Smart Dustbin                       | Ultrasonic sensor, servo             |
| 22  | Automatic Plant Irrigation          | Moisture sensor, pump, relay         |
| 23  | Smart Parking System                | IR/ultrasonic sensors, display       |
| 24  | Bluetooth Home Automation           | ESP32/HC-05, relay module            |
| 25  | Wi-Fi Home Automation               | ESP32/ESP8266, MQTT/web control      |
| 26  | IoT Energy Meter                    | Current/voltage sensing, ESP32       |
| 27  | Digital Energy Monitoring Dashboard | ESP32, cloud dashboard               |
| 28  | Smart LPG Cylinder Monitor          | Load cell + gas sensor + alert       |

| No. | Project                        | Typical implementation / key modules |
|-----|--------------------------------|--------------------------------------|
| 29  | Gas Leakage Auto Shutoff       | Gas sensor + solenoid valve          |
| 30  | Smart Water Tank Controller    | Level sensing + pump automation      |
| 31  | RFID Attendance System         | RFID + RTC + storage                 |
| 32  | Face-Recognition Door Lock     | ESP32-CAM/Raspberry Pi               |
| 33  | Fingerprint Security System    | Fingerprint module + relay           |
| 34  | Smart Helmet Safety System     | Helmet sensor + alcohol sensor + GPS |
| 35  | Accident Detection & Alert     | IMU + GPS + GSM/IoT                  |
| 36  | Vehicle Anti-Theft System      | GPS/GSM + vibration sensor           |
| 37  | Blind-Spot / Parking Assist    | Ultrasonic sensors + buzzer          |
| 38  | Automatic Fan Speed Controller | Temperature sensor + PWM             |
| 39  | Digital Thermostat             | Temperature sensing + relay          |
| 40  | Solar Panel Tracking System    | LDRs + servo/actuator                |

### 3. IoT, AI & Smart-System Projects

| No. | Project                           | Typical implementation / key modules  |
|-----|-----------------------------------|---------------------------------------|
| 41  | IoT Weather Station               | ESP32 + multi-sensor node             |
| 42  | Cloud-Based Air Quality Monitor   | PM2.5 + gas sensors + dashboard       |
| 43  | Smart Agriculture Monitoring      | Soil, temperature, humidity, light    |
| 44  | Crop Disease Detection            | Camera + ML image classification      |
| 45  | Plant Health Monitoring System    | Sensors + IoT analytics               |
| 46  | Smart Waste Segregation           | Sensors/camera + servo mechanisms     |
| 47  | AI-Based Fire Detection           | Camera + computer vision              |
| 48  | IoT Fire & Smoke Alert System     | Smoke/temp sensor + cloud alert       |
| 49  | Smart Streetlight with IoT        | LDR/PIR + remote monitoring           |
| 50  | Smart Building Energy Management  | Sensors + load analytics              |
| 51  | Predictive Maintenance Monitor    | Vibration + temperature + ML          |
| 52  | Industrial Machine Health Monitor | Current/vibration/temperature sensing |
| 53  | IoT-Based Water Quality Monitor   | pH/TDS/turbidity/temp                 |
| 54  | Smart Flood Warning System        | Water level + rain + IoT              |
| 55  | Remote Patient Room Monitoring    | Non-invasive environmental sensing    |

### 4. Advanced Electronics, Semiconductor & EMT Projects

| No. | Project                                  | Typical implementation / key modules |
|-----|------------------------------------------|--------------------------------------|
| 56  | MOSFET-Based Electronic Load             | Power MOSFET + current sensing       |
| 57  | DC-DC Buck Converter Design              | MOSFET switching + feedback          |
| 58  | Boost Converter with Closed-Loop Control | PWM + feedback controller            |
| 59  | SPWM Inverter                            | MCU + MOSFET bridge + driver         |
| 60  | BLDC Motor Controller                    | Hall sensors + MOSFET bridge         |
| 61  | Three-Phase Inverter                     | Gate driver + power MOSFET/IGBT      |

| No. | Project                                      | Typical implementation / key modules |
|-----|----------------------------------------------|--------------------------------------|
| 62  | Power Factor Correction Prototype            | Current/voltage sensing + control    |
| 63  | Battery Management System                    | Cell monitoring + balancing          |
| 64  | Lithium Battery Protection System            | Protection IC + MOSFETs              |
| 65  | Solar MPPT Charge Controller                 | DC-DC converter + MPPT algorithm     |
| 66  | Digital Oscilloscope Prototype               | ADC + MCU + display                  |
| 67  | Logic Analyzer                               | High-speed sampling + USB/MCU        |
| 68  | Function Generator                           | DDS/DAC + waveform control           |
| 69  | PLL Frequency Synthesizer                    | Phase detector + divider + VCO       |
| 70  | Digital Frequency Meter                      | Timer/counter + display              |
| 71  | RF Signal Strength Meter                     | RF detector + ADC                    |
| 72  | EMI/EMC Noise Monitoring Device              | Probe + detector + spectrum analysis |
| 73  | PCB Transmission-Line Characterization       | S-parameter/impedance measurement    |
| 74  | Antenna VSWR / Return-Loss Meter             | RF detector + directional coupler    |
| 75  | Temperature-Controlled Semiconductor Chamber | Sensor + PID + heater/cooling        |
| 76  | Semiconductor Wafer Temperature Monitor      | Thermocouple/RTD + DAQ               |
| 77  | Photodiode Characterization System           | IV measurement + data logging        |
| 78  | Solar Cell IV Curve Tracer                   | Programmable load + ADC              |
| 79  | LED IV & Optical Power Characterization      | Current sweep + photodiode           |
| 80  | Thermistor / RTD Characterization            | Resistance-temperature measurement   |

## 5. Advanced Communication, Robotics & Control Projects

| No. | Project                             | Typical implementation / key modules |
|-----|-------------------------------------|--------------------------------------|
| 81  | LoRa-Based Environmental Monitoring | LoRa nodes + gateway                 |
| 82  | BLE-Based Smart Sensor Network      | ESP32/BLE mesh concepts              |
| 83  | Wi-Fi Intrusion Alert System        | ESP32 + sensors + notifications      |
| 84  | RFID-Based Smart Locker             | RFID + motor/solenoid + logging      |
| 85  | Line-Following Autonomous Robot     | IR array + motor control             |
| 86  | Obstacle-Avoiding Robot             | Ultrasonic + motor driver            |
| 87  | Bluetooth-Controlled Robot          | BLE/Bluetooth + motor driver         |
| 88  | Self-Balancing Robot                | IMU + PID control                    |
| 89  | Robotic Arm with Object Sorting     | Servo control + sensors/camera       |
| 90  | Autonomous Fire-Fighting Robot      | Flame sensing + navigation           |
| 91  | PID Motor Speed Controller          | Encoder + PWM + feedback             |
| 92  | DC Motor Condition Monitoring       | Current + vibration + temperature    |
| 93  | CAN Bus Vehicle Monitor             | CAN transceiver + MCU                |
| 94  | Modbus RTU Industrial Monitor       | RS-485 + MCU                         |
| 95  | Industrial IoT Gateway              | RS-485/CAN + Ethernet/Wi-Fi          |

## Project Selection Guide

**For quick 1-day builds:** choose sensor, alarm, RFID, temperature, rain, moisture, ultrasonic or automatic-light projects. **For stronger final-year projects:** choose IoT monitoring, smart agriculture, energy monitoring, predictive maintenance, BMS, MPPT, BLDC control, CAN/Modbus or industrial gateways. **For semiconductor/EMT-oriented work:** choose MOSFET converters, inverter systems, PLLs, RF measurement, EMI/EMC monitoring, photodiode/solar-cell characterization, or semiconductor thermal characterization. Verify component availability and project requirements before promising a delivery timeline.

Contact: 9663076094 | Pricing: Project-dependent and negotiable | Fast delivery / student-friendly options